Verified Recursive Self-Training and Knowledge Distillation for Amharic News Topic Classification

iCog Labs AI Internship: Emerging AI Paradigms assignment

1 Setup

Data. Amharic MasakhaNEWS (official splits): 1,311 train, 188 dev, 376 test; four classes (business, health, politics, sports). Text is headline plus body, truncated to 3,000 characters. The dataset license was not verified. **Protocol.** Tasks 1 and 3 use a stratified labelled seed set of 20 articles per class (80 labels); the other 1,231 training articles are an unlabeled pool whose gold labels are used only to measure pseudo-label precision. Dev decides accept/rollback and (T, α) selection; test is used only for final reporting. All numbers are mean ± std over 3 seeds (0,1,2). **Models.** Teacher: 2-layer MLP (256 hidden) on word 1–2-gram TF-IDF (10k features). Student: linear model on a 2k-feature unigram TF-IDF view (8,004 parameters). Independent verifier and baseline: logistic regression on character 2–4-gram TF-IDF. AfroXLMR features are implemented but were not run.

2 Task 1: verified recursive self-training

Each round pseudo-labels the pool, accepts labels with confidence ≥ 0.9 (verified variant: and agreement with the independent character model), retrains a copy on seed plus accepted examples, and accepts the copy only if dev macro-F1 does not drop; two consecutive rejections stop the loop (at most 5 rounds).

Model	Test macro-F1	Test ECE
Char-LR, seed only	0.873 ± 0.018	0.400 ± 0.018
Round-0 teacher (seed only)	0.848 ± 0.015	0.304 ± 0.017
Verified loop, final teacher	0.845 ± 0.011	0.253 ± 0.008
Confidence-only loop, final	0.845 ± 0.011	0.253 ± 0.008
Naive loop (no rollback), final	0.708 ± 0.062	0.068 ± 0.009
Teacher, all gold labels (ref.)	0.910 ± 0.006	0.037 ± 0.010

The verified loop gave no reliable test gain (per-seed change in test macro-F1: +0.005, −0.003, −0.017), although dev improved after round 1. The agreement check changed nothing relative to confidence-only. Without rollback, test macro-F1 fell by 0.11–0.17 even though pseudo-label precision stayed at 0.97–1.00; the entropy of accepted classes decreased across rounds, consistent with class drift toward easy classes (class counts were not inspected directly). The character-LR baseline trained on the seed set alone scored above the MLP teacher.

3 Task 2: knowledge distillation

Loss: $\alpha\,\mathrm{CE}(y, s) + (1-\alpha)\,T^2\,\mathrm{KL}\big(\mathrm{softmax}(t/T)\,\|\,\mathrm{softmax}(s/T)\big)$; sweep $T \in \{1, 2, 4\}$, $\alpha \in \{0.3, 0.5, 0.7\}$, selected on dev. Teacher trained on all gold train labels.

Model	Test macro-F1	Test ECE	Params	CPU ms/example
Teacher (MLP)	0.910 ± 0.006	0.037 ± 0.010	2,561,284	0.1090 ± 0.0122
Student, hard labels	0.901 ± 0.000	0.065 ± 0.004	8,004	0.0269 ± 0.0007
Student, distilled (T=4, α=0.5)	0.902 ± 0.004	0.039 ± 0.009	8,004	0.0261 ± 0.0033
Char-LR baseline	0.925 ± 0.000	0.066 ± 0.000	200,004	–

The student has about 320× fewer parameters and runs about 3.4× faster than the teacher at a cost of roughly 0.7 to 0.9 macro-F1 points. Distillation did not improve student accuracy over hard labels; it lowered test ECE. The character-LR baseline was the strongest model here but is much larger (200,004 parameters).

4 Task 3: merged pipeline

The Task 1 loop produces a teacher that is distilled into the student using $(T, \alpha) = (4, 0.5)$ from Task 2. Hard labels exist only for the seed set during distillation.

Model	Test macro-F1	Test ECE
teacher_round0	0.848 ± 0.015	0.304 ± 0.017
teacher_final_loop	0.845 ± 0.011	0.253 ± 0.008
student_seed_hard	0.830 ± 0.022	0.292 ± 0.019
student_hard_pseudo	0.818 ± 0.024	0.247 ± 0.025
student_from_round0	0.810 ± 0.032	0.246 ± 0.028
student_from_final	0.816 ± 0.035	0.214 ± 0.021
student_task2_full_labels(ref.)	0.902 ± 0.004	0.039 ± 0.009
teacher_full_labels(ref.)	0.910 ± 0.006	0.037 ± 0.010

Mean change (final minus round-0): teacher test -0.0037, student test +0.0060, teacher dev +0.0214, student dev +0.0061. Because the teacher gain on test is itself within noise, these results do not show that the loop's improvement survives distillation; they are a null result with 3 seeds.

5 Limitations

Three seeds and a 188-example dev set make accept/rollback decisions noisy. The teacher is underconfident at small label budgets (high ECE), which limits how many pseudo-labels pass the 0.9 threshold. The verifier is a proxy rather than ground truth. TF-IDF features were used throughout, so conclusions may differ with an AfroXLMR encoder. Dataset license unverified.